

Marios Papachristou

Assistant Professor of Information Systems
W. P. Carey School of Business
Arizona State University

LAST UPDATED	September 8, 2026
PERSONAL INFORMATION	E-mail mpapachr@asu.edu Website papachristoumarios.github.io Google Scholar Profile Google Scholar Profile Campus Address 400 E Lemon St, BAC, Tempe, AZ 85281 US Status US Permanent Resident (Green Card Holder)
RESEARCH	My research develops models for understanding and improving the <i>resilience</i> of modern socio-technical systems with downstream applications to platforms, supply chains, and organizations. I study the mechanisms causing fragility and devise analytical metrics to measure the resilience, prescriptive interventions, and decentralized decision-making under uncertainty and information constraints.
METHODS	Network Science, Algorithms, AI Agents, Privacy, Computational Social Science, ML
ACADEMIC APPOINTMENTS	W. P. Carey School of Business, Arizona State University 2025 – Tenure-Track Assistant Professor, Department of Information Systems <i>Affiliated Member</i>, Center for AI and Data Analytics (AIDA) <i>Graduate Faculty Member</i>, Data Science, Analytics and Engineering <i>Graduate Faculty Member</i>, Computer Science Cornell University Summer 2026 Visiting Faculty
INDUSTRY EMPLOYMENT	Microsoft Research Summer 2023 Office of Applied Research, <i>with Chin-Chia Hsu and Longqi Yang</i> Twitter Summer 2022 User Modeling Research Group, <i>with Rishab Goel, Frank Portman, Matt Miller</i>
EDUCATION	Cornell University 2020 – 2025 Ph.D. in Computer Science, Minor: <i>Applied Mathematics</i> <i>Doctoral Advisor</i>: Jon Kleinberg <i>Thesis Title</i>: “From Contagion to Stability: Insights into Network Dynamics, Resilience and Stability”. Nominated for the <i>ACM Doctoral Dissertation Award</i> M.S. in Computer Science 2020 – 2022 National Technical University of Athens 2015 – 2020 Diploma in Electrical and Computer Engineering <i>Advisor</i>: Dimitris Fotakis
HONORS & AWARDS	– INFORMS eBusiness Section Best Student Paper Award 2026 Finalist (best paper or runner-up). – ACM Doctoral Dissertation Award Nominee 2025 Nominated to the ACM by Cornell’s CS department as one of the two best Ph.D. theses in engineering-related disciplines awarded in 2025.

- **Sigma Xi** 2025
Elected member based on scientific contributions.
- **Cornell ORIE Young Researchers Workshop** 2024
Invited workshop.
- **Alexander A. Onassis Scholarship** 2023
Three-year fellowship for Greek Ph.D. students. Acceptance rate: 16%.
- **Bodossaki Foundation Fellowship** 2023
Runner-up.
- **LinkedIn Ph.D. Fellowship** 2022
Awarded one-year fully funded fellowship by LinkedIn. Acceptance rate 14%.
- **A.G. Leventis Scholarship** 2022
Award for graduate students of Greek origin. Acceptance rate 12.5%.
- **Gerondelis Scholarship** 2022
Awarded one-time award for graduate students of Greek descent.
- **Chateaubriand Fellowship** 2022
Awarded fellowship granted by the French Embassy in the US for studies in France (declined). Acceptance rate 33%.
- **Cornell Fellowship** 2020
Awarded one-year fully funded fellowship by the CS department based on “exceptional preparation and promise”.
- **Thomaidion Award** 2019
Awarded by NTUA for publication during undergraduate studies.
- **ESEC/FSE 2019 ACM Student Research Competition Finalist** 2019
- **CanSat International Space Engineering Competition** 2019
Awarded 4th out of 93 teams. Organized by the American Astronautical Society (AAS).
- **2nd prize “be finnovative 2.0” Start-up Accelerator by National Bank of Greece** 2018
Awarded 2nd prize for our cashierless payments start-up.
- **1st prize at “Crowdhackathon Fintech #2” by the National Bank of Greece** 2017
Awarded 1st prize for our cashierless payments start-up.
- **IEEEExtreme 11.0 Programming Competition** 2017
Awarded top 1% worldwide in IEEE’s algorithms competition.
- **Touramanoglu Scholarship** 2015
Awarded for the highest University Entrance Exams score in the Imittos region.
- **“The Great Moment of Education” Scholarship by Eurobank SA** 2015
Awarded for the highest University Entrance Exams score in the Imittos region.

JOURNAL PAPERS

$\alpha\beta$ = alphabetical order * = equal contribution

$^{\circ}$ = corresponding author ‡ = graduate student supervised

[J1] Network Formation and Dynamics among Multi-LLMs

Marios Papachristou, Yuan Yuan

PNAS Nexus, 2025

[\[paper\]](#) [\[preprint\]](#) [\[code\]](#)

– Taught at UC Berkeley, Stanford, UIUC and others

– Featured by *TechXplore*, and *SingularityHub*

[J2] Leveraging Large Language Models for Collective Decision-Making

Marios Papachristou, Longqi Yang, Chin-Chia Hsu

Proceedings of the ACM on Human Computer Interaction (presented at ACM CSCW '25), 2025

[\[paper\]](#)

- Invited talk at the LLMs as Research Tools Workshop at ACM CHI 2024
- Featured by the *ACM*

- [J3] **volesti: A C++ library for sampling and volume computation on convex bodies**
 ^{$\alpha\beta$} Apostolos Chalkis*, Vissarion Fisikopoulos*, *Marios Papachristou**, Elias Tsigaridas*
Journal of Open Source Software, 2025
[\[paper\]](#) [\[code\]](#)
- [J4] **Differentially Private Distributed Estimation and Learning**
Marios Papachristou, Amin Rahimian
IJSE Transactions (formerly IIE Transactions), 2025
[\[paper\]](#) [\[code\]](#) [\[talk\]](#)
 - Featured at the *July 2025 issue of the ISE Magazine* (10% of articles published in IJSE Transactions)
 - Featured by the *University of Pittsburgh Swanson School of Engineering*
- [J5] **Truncated Log-concave Sampling for Convex Bodies with Reflective Hamiltonian Monte Carlo**
 ^{$\alpha\beta$} Apostolos Chalkis, Vissarion Fisikopoulos, *Marios Papachristou*^o, Elias Tsigaridas
ACM Transactions on Mathematical Software (TOMS), 2023
[\[paper\]](#) [\[code\]](#)
 - Featured at *ODBMS.org*
- [J6] **Sublinear Domination and Core-periphery Networks**
Marios Papachristou (solo-authored)
Scientific Reports, 2021
[\[paper\]](#) [\[code\]](#)
- [J7] **Commons-based peer production and digital fabrication: The case of a RepRap-based, Lego-built 3D printing-milling machine**
Vasilis Kostakis*, and *Marios Papachristou**
Telematics and Informatics, 2014
[\[paper\]](#)
- TOP-TIER CS
CONFERENCE
PROCEEDINGS
- [C1] **Opinion Dynamics with Multiple Adversaries**
Akhil Jalan*, *Marios Papachristou*^{*o}
WWW '26: *ACM Web Conference (WWW)*, 2026
[\[paper\]](#)
 - Acceptance rate: 20%
- [C2] **Dynamic Interventions for Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
WWW '23: *ACM Web Conference (WWW)*, 2023
[\[paper\]](#) [\[code\]](#)
 - Acceptance rate: 17%
- [C3] **Core-periphery Models for Hypergraphs**
Marios Papachristou, Jon Kleinberg
KDD '22: *ACM Intl. Conf. on Knowledge Discovery and Data Mining (KDD)*, 2022
[\[paper\]](#) [\[code\]](#) [\[data\]](#)
 - Acceptance rate: 15%
 - Taught at tutorials at KDD and CIKM
- [C4] **Allocating Stimulus Checks in Times of Crisis**
Marios Papachristou, Jon Kleinberg
WWW '22: *ACM Web Conference (WWW)*, 2022
[\[paper\]](#) [\[code\]](#)
 - Acceptance rate: 17%
 - Featured by the *Federal Reserve Bank of Minneapolis*
 - Taught at UC Berkeley

- [C5] **Software Clusterings with Vector Semantics and the Call Graph**
Marios Papachristou (solo-authored)
ESEC/FSE '19: ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), 2019
[\[paper\]](#) [\[code\]](#) [\[data\]](#)
– Acceptance rate: 21%
– ESEC/FSE Student Research Competition Finalist Paper

REVISIONS

- [WP1] **The Power of Choice in Random Sampling and Applications to Platform Analytics**
Marios Papachristou, Jon Kleinberg
Major revision, Operations Research, 2025
[\[preprint\]](#)
– Taught at the University of Pittsburgh
- [WP2] **Structural Measures of Resilience for Supply Chains**
Marios Papachristou, Amin Rahimian, Arash Azadegan
Major revision, Production and Operations Management, 2025
[\[preprint\]](#) [\[code\]](#)
- [WP3] **Echoes of Disagreement: Measuring Disparity in Social Consensus**
Marios Papachristou, Jon Kleinberg
Major revision, INFORMS Journal of Computing, 2025
[\[preprint\]](#) [\[code\]](#)

WORKING
PAPERS

- [WP4] **Data Regulation and the Microstructure of Clinical Trials**
Kyuyeon (Jenny) Rhee[‡], *Marios Papachristou*, Tian Lu, Pei-yu Chen
Under preparation for submission, 2026
Preprint available upon request
– *INFORMS eBusiness Best Paper Award Finalist (best paper or runner-up)*
- [WP5] **Differentially Private Distributed Inference for Multicenter Clinical Studies**
Marios Papachristou, Amin Rahimian
Submitted, 2026
Preliminary version presented at the Theory and Practice of Differential Privacy Workshop (TPDP)
[\[preprint\]](#)
- [WP6] **The Price of Misreporting: Measuring the Effect of Strategic Behavior in Online Platforms**
Marios Papachristou, Akhil Jalan
Submitted, 2026
Preliminary version presented at the ACM Web Conference 2026
[\[preprint\]](#)
- [WP7] **Optimal Resource Allocation for Remediating Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
Submitted, 2026
[\[preprint\]](#)
- [WP8] **From Thick Description To Computational Simulation: An Agentic Framework For Deriving Simulation Studies From Qualitative Research**
Julian Lehmann, David Dobolyi, *Marios Papachristou*, Gerald Kane
Submitted, 2026
Preprint available upon request

WORKS IN PROGRESS	[WIP1] Impact of Massive AI-generated Content on Recommender Systems with Joonhwa Park[‡], and Panos Adamopoulos [WIP2] The Effects of Generative AI on Consumer Behavior with Panos Adamopoulos
PUBLISHED SOFTWARE	volesti github.com/GeomScale/volesti Open-source C++ statistical software for sampling and Bayesian inference.
REFEREED CS WORKSHOPS	[W1] GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs <i>Marios Papachristou</i>, Rishab Goel, Frank Portman, Matt Miller, Rong Jin <i>NeurIPS workshop on Graph Learning Frontiers (GLFrontiers)</i>, 2022 [paper] [poster]
TUTORIALS	[T1] Differential Privacy for Strategic Information Sharing and Learning: Foundations, Mechanisms, and Applications Amin Rahimian, Juba Ziani, <i>Marios Papachristou</i>, Yuxin Liu <i>Conference on Web and Internet Economics (WINE)</i>, 2025
TEACHING	Arizona State University, Instructor <i>Spring 2026</i> CIS 509: Analytics for Unstructured Data – Evaluation (out of 7.0, MSBA/MSBA/MSAIB): 6.8/7.0, 6.6/7.0, 6.4/7.0 (Spring 2026), Department Average: 6.2/7.0 Cornell University, Teaching Assistant <i>Spring 2023, Fall 2023</i> CS 6850: The Structure of Information Networks (Ph.D.-level) – Evaluation (out of 5.0): 5.0/5.0 (Spring 2023), 4.8/5.0 (Fall 2023)
GRANTS	– ALCF Director’s Discretionary Allocation Award (DD Award) <i>2026</i> Argonne Leadership Computing Facility. Project: Large-scale LLM-based Simulations for Algorithmic Auditing of Recommendation Systems
MENTEES	– Kyuyeon (Jenny) Rhee (Ph.D. student at ASU) – Joonhwa Park (Ph.D. student at Emory) – Victor Lee (Ph.D. student at ASU) – Pin-Jun Liu (Ph.D. student at ASU) – Jasmine Li (Undergraduate Cornell CS) – Ioannis Iakovidis (Google Summer of Code 2022; now Ph.D. student at Johns Hopkins) – Zhang Zhuyan (Google Summer of Code 2022; now Technology analyst at Bank of America)
MENTORSHIP EXPERIENCE	– Student-applicant Support Program at Cornell CIS <i>2022</i> – Google Summer of Code Mentor <i>2019 – 2022</i> – Volunteer lectures for the Greek National Olympiad of Informatics (PDP) <i>2019</i>
INVITED TALKS & LECTURES [†] = scheduled [*] = presented by co-author	• The Price of Misreporting: Measuring the Effect of Strategic Behavior in Online Platforms – Eller College of Management, University of Arizona[†] <i>November 2026</i> – Lerner-Warwick FinTech Workshop[†] <i>November 2026</i> – NYU Stern School of Business, WAICI Workshop[†] <i>September 2026</i> • Resilience in Networks: Frameworks for Resource Allocation and Simulation – Archimedes, Athena RC <i>April 2025</i>

- TOC4Fairness Seminar, Simons Institute *March 2025*
- Cornell University, Theory Seminar *December 2024*
- Arizona State University, W. P. Carey School of Business *November 2024*
- Purdue University, Daniels School of Business *November 2024*
- MIT, Research Group Meeting *June 2024*
- ***Network Formation and Dynamics among Multi-LLMs***
 - Arizona State University, School of Mathematical and Natural Sciences, AI Seminar *November 2025*
 - Arizona State University, Machine Learning Day *April 2025*
 - IBM Research, Exploratory Math Sciences Council Seminar *April 2024*
 - UIUC, Guest Lecture at the Network Analysis Course *April 2024*
- ***Information Aggregation and Distributed Learning in Privacy-critical Environments***
 - CMU, Tepper School of Business OM Seminar* *September 2024*
 - TOC4Fairness Online Seminar Series* *March 2024*
 - Rutgers Business School, Seminar* *November 2023*
- ***Leveraging Large Language Models for Collective Decision-Making***
 - LLMs as Research Tools: Applications and Evaluations in HCI Data Work Workshop Invited Talk, ACM Conference on Human-Computer Interaction* *May 2024*
 - Microsoft Research, Invited Applied Research Talks *August 2023*
- ***Structural Measures of Resilience for Supply Chains***
 - Columbia University, Theory Seminar *November 2023*
- ***Resource Allocation in a Financial Contagion Environment***
 - University of Chicago, Theory Seminar *October 2023*
 - Indiana University, Center for Complex Networks and Systems Research *September 2023*
 - Cornell University, CS 6850 Guest Lecture *April 2023*
 - Cornell University, Theory Seminar *November 2022*
- ***GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs***
 - Twitter, Machine Learning Seminar *August 2022*

DEPARTMENTAL
AND
PROFESSIONAL
SERVICE

Department and School Outreach

- AI and Data Analytics Center (AIDA) Brown Bag Seminar Series Co-organizer [\[link\]](#)
- Department of Information Systems Brown Bag Seminar Co-organizer [\[link\]](#)
- W. P. Carey School of Business, PhD Lunch and Learn, November 2025, Speaker [\[link\]](#)

Program Committee

- ACM Conference on Fairness Accountability and Transparency (FAccT)
- ACM Conference on Knowledge Discovery and Data Mining (KDD)
- ACM Conference on Economics and Computation (EC)

Ad-hoc Reviewer

- Management Science
- MIS Quarterly
- INFORMS Journal of Computing
- Mathematics of Operation Research
- ACM Transactions on Management Information Systems
- Journal of Artificial Intelligence Research
- Naval Research Logistics
- International Conference on Computational Social Science (IC2S2)

- ACM Web Conference (WWW)
- Innovations in Theoretical Computer Science (ITCS)
- ACM Conference on Human Factors in Computing (CHI)
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)
- Machine Learning
- NeurIPS 2022 Workshop on Graph Learning Frontiers (GLFrontiers)
- NeurIPS 2021 Workshop on Human and Machine Decisions
- Journal of Open Source Software
- IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS)

Session Chair

- INFORMS Annual Meeting 2025
- NSF Conference on Network Science and Economics 2024
- ACM Web Conference (WWW) 2022

REFERENCES

Available upon request.